E. Vlad and a Pair of Numbers

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: bitmasks, constructive algorithms | https://codeforces.com/problemset/problem/1790/E

Vlad found two positive numbers a and b (a,b gt 0). He discovered that $a \oplus b = (a + b)/2$, where $\oplus$ means the bitwise exclusive OR , and division is performed without rounding..

Since it is easier to remember one number than two, Vlad remembered only $a \oplus b$, let's denote this number as x. Help him find any suitable a and b or tell him that they do not exist.

Input

The first line of the input data contains the single integer t ($1 \le t \le 10^4$) — the number of test cases in the test.

Each test case is described by a single integer x ($1 \le x \le 2^{29}$) — the number that Vlad remembered.

Output

Output t lines, each of which is the answer to the corresponding test case. As the answer, output a and b (0 lt a,$b \le 2^{32}$), such that $x = a \oplus b = (a + b)/2$. If there are several answers, output any of them. If there are no matching pairs, output -1.

Example

Input
6 2 5 10 6 18 36

Output
3 1 -1 13 7 -1 25 11 50 22